

Project Executable File

This document lists all executable files, scripts, and runnable components of the OptiCrop project, along with their purpose and usage instructions.

Primary Executable Flask Web Application

Property	Details
File	app.py (project root)
Runtime	Python 3.11 (venv required)
Command	python app.py
Default URL	http://127.0.0.1:5000
Dependencies	All listed in requirements.txt install via pip install -r requirements.txt
Pre-requisite	Run python scripts/train_model.py first to generate models/best_model.pkl

Setup & Run Step by Step

Step 1: Clone the repository

```
git clone https://github.com//opticrop.git
```

Step 2: Navigate to project folder

```
cd opticrop
```

Step 3: Create virtual environment

```
python -m venv venv
```

Step 4: Activate virtual environment

```
venv\Scripts\activate (Windows) | source venv/bin/activate (Linux/macOS)
```

Step 5: Install dependencies

```
pip install -r requirements.txt
```

Step 6: Train models (first time only)

```
python scripts/train_model.py
```

Step 7: Start the Flask application

```
python app.py
```

Step 8: Open in browser

```
http://127.0.0.1:5000
```

requirements.txt Key Dependencies

Package	Version	Role
Flask	3.0.3	Web framework routing, templating, request handling

scikit-learn	1.4.2	ML algorithms Random Forest, KNN, Decision Tree, Logistic Regression, K-Means
pandas	2.1.4	Data loading and manipulation
numpy	1.26.4	Numerical operations and array handling
matplotlib	3.8.4	Model evaluation plots in Jupyter notebooks
seaborn	0.13.2	Statistical visualisation in EDA notebooks
gunicorn	22.0.0	WSGI server for production deployment
pytest	8.2.0	Unit and integration testing framework